

SAR 1B - EO Data Quality & Processing Validation for Agricultural Monitoring in Emilia-Romagna

Extension of SAR 1. Study area: Imola-Castel Bolognese-Faenza corridor, Emilia-Romagna, Italy.

Purpose

SAR 1B converts the existing agricultural Sentinel-1/Sentinel-2 workflow into an explicit Earth Observation data-quality and processor-validation chain:

dataset -> quality control -> anomaly investigation -> reprocessing -> validation -> reporting

The workflow is intentionally traceable. Thresholds are project QC rules documented in metadata/SAR1B_QC_THRESHOLDS.json; they are not represented as ESA mission acceptance specifications.

Executive QC result

- Final register: **73 PASS, 3 WARNING, 0 FAIL.**
- Sentinel-1 acquisition series: **30** path-95/frame-445 descending VV+VH acquisitions from 2023-01-10 to 2023-12-24.
- RTC QA: **60** polarization rasters; minimum valid fraction **100.000%**; no acquisition-wide anomaly flag.
- SLC/coherence: **6** 12-day pairs. One geometry warning is retained for the February pair because $|\text{baseline}| = 225.6 \text{ m}$, above the project 200 m investigation band.
- Sentinel-2: **5** dates. The 2023-10-10 acquisition is retained as **WARNING** because only **89.4%** of pilot polygons reach 80% valid support and some polygons have zero valid pixels.
- S1 x S2 temporal matching: mean offset **2.6 d**, maximum **4 d**; the 4-day July match is a **WARNING** under the project $\leq 3 \text{ d}$ **PASS** rule.
- Automated parcel-level anomaly detector generated **54** **INVESTIGATE** triggers, but **0** acquisition-wide anomaly flags. Parcel anomalies are therefore not automatically treated as product failures.

Controlled reprocessing / processor verification

The Sentinel-2 index processor was independently re-run from the stored band mosaics and SCL layer using the original scale/offset, cloud-mask logic and bilinear alignment of 20 m bands. Pixelwise comparison with the previously generated products gives a maximum replication RMSE of **0.000e+00**, verifying reproducibility of the baseline processor.

Two controlled parameter changes were then tested:

1. **Strict SCL mask:** adds SCL classes 2 (dark area pixels) and 7 (unclassified) to the exclusion policy while retaining the original bilinear resampling.
2. **20 m resampling variant:** changes B05/B11 alignment from bilinear to nearest-neighbour while retaining the original SCL mask.

The strict-mask experiment produced a worst parcel-level relative valid-pixel change of **0.0%**. The largest 95th-percentile absolute polygon-median change produced by the nearest-neighbour resampling experiment was **0.0099** index units. These results quantify the effect of processing choices rather than assuming processor invariance.

PASS / WARNING / FAIL items

product_type	product_id	date_or_pair	status	reason
S1_COHERENCE	20230203_20230215	20230203_20230215	WARNING	
S2_L2A	20231010	2023-10-10	WARNING	89.4% polygons >=80% valid; minimum=0.
S1_S2_PAIR	20230721_20230717	2023-07-21 / 2023-07-17	WARNING	temporal offset=4 d; native CRS differs (S1 EPSG:32633 \ S2 EPSG:32632); comparison uses common polygon support

QA dimensions implemented

Sentinel-1 acquisitions

Relative orbit, frame, flight direction, VV/VH availability, acquisition cadence and temporal gaps are checked against the project configuration.

Sentinel-1 RTC

The workflow checks raster presence, CRS/resolution, positive finite gamma0 support, no-data fraction, edge-versus-interior validity and robust temporal distribution anomalies for VV and VH.

SLC/coherence

Each pair is checked for temporal interval, perpendicular baseline, looks, output resolution, polygon coverage and physical coherence range. Baseline-driven warnings are separated from seasonal coherence variability.

Sentinel-2

SCL-based valid support is measured per pixel polygon together with cloud, shadow, saturated/defective, no-data, dark-area and unclassified fractions. Availability is summarized at acquisition level.

Cross-sensor comparability

Nearest-date S1/S2 matches, temporal offsets, native CRS, nominal resolution and common AOI footprint are reported. S1 is EPSG:32633 and S2 is EPSG:32632 in the stored products; analyses therefore use common polygon support rather than assuming pixel-grid identity.

Automated anomaly investigation

Robust MAD-based detectors operate on S1 temporal parcel series and on within-class S2/coherence distributions. Statistical anomalies are labelled INVESTIGATE and only escalate acquisition status if they become widespread.

Interpretation

The SAR 1B workflow does not claim to reproduce an ESA operational IPF or mission-specific acceptance procedure. It demonstrates the transferable engineering logic required for instrument/product-quality support: systematic QC, traceable status assignment, anomaly isolation, controlled reprocessing, quantitative comparison, validation and reporting using real EO products.

Reproducible outputs

- code/40_sar1b_dataset_qc.py
- code/41_sar1b_anomaly_detection.py
- code/42_sar1b_reprocessing_validation.py
- code/43_sar1b_reporting.py
- data/processed/qc/product_qc_register.csv
- data/processed/qc/qc_report.json

- outputs/figures_qc/15_qc_availability_timeline.png through 20_qc_status_summary.png

[executed on device: DESKTOP-OV1FMCI (0e52685a-65a1-435c-b5c2-28d4e3be5d0b)]

QC Figures

15 qc availability timeline



16 rtc distribution qc



17 s2 valid fraction qc



18 coherence pair qc



19 reprocessing impact



20 qc status summary

